

AI adoption, productivity and employment: Evidence from European firms

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Abstract

This paper provides new evidence on how the adoption of artificial intelligence (AI) affects productivity and employment in Europe. Using matched EIBIS-ORBIS data on more than 12,000 non-financial firms in the European Union (EU) and United States (US), we instrument the adoption of AI by EU firms by assigning the adoption rates of US peers to isolate exogenous technological exposure. Our results show that AI adoption increases the level of labor productivity by 4%. Productivity gains are due to capital deepening, as we find no adverse effects on firm-level employment. This suggests that AI increases worker output rather than replacing labor in the short run, though longer-term effects remain uncertain. However, productivity benefits of AI adoption are unevenly distributed and concentrate in medium and large firms. Moreover, AI-adopting firms are more innovative and their workers earn higher wages. Our analysis also highlights the critical role of complementary investments in software and data or workforce training to fully unlock the productivity gains of AI adoption.

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